

JahnaviBadam

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SKILLS

Programming Languages: Python, C++, Java, JavaScript.

Web Technologies: HTML, CSS (with Bootstrap), ReactJs, Spring Boot, NodeJs.

Databases: MySQL, MongoDB and PostgreSQL.

IT Constructs: Data Structures and Algorithms, OOPs, Operating Systems, DBMS and Computer Networks. Tableau, Git, MATLAB, Simulink.

PROFESSIONAL EXPERIENCE

Software Engineer
NatWest Group

07/2023 – present

- Built a scalable **Resource Management Tool** using **ReactJS** and **Spring Boot** to track team resources, project portfolios, and allocations. Worked in a team of 3, improving resource efficiency by 30% for 300+ employees. Automated workflows, cutting manual effort by 25% and enhancing operational productivity. Implemented real-time data visualization and database integration for seamless tracking and reporting.
- Played a key role in upgrading **Splunk Universal Forwarder** across 200+ test and production servers to version 9.0.3, ensuring 100% compliance with platform security and performance standards. Improved log ingestion efficiency by 20%, minimizing system downtime and enhancing monitoring reliability.
- Designed and optimized interactive **Tableau dashboards** for real-time data visualization, streamlining production-level reporting. Reduced report generation time by 30%, enabling faster insights and data-driven decision-making for a 50+ member team.

PROJECTS

Maximizing sum rate in a wireless communication channel using distributed DNNs

- Developed a **deep learning-based power control model** for wireless interference channels with imperfect channel state information (CSI), using **Python**(Tensorflow, Keras, NumPy) achieving up to 95% of optimal sum spectral efficiency while mitigating performance degradation of 5% to 20% due to CSI errors, using an ensemble of 5 neural networks trained over 4,000 epochs.

Implementation of an AutoEncoder using PyTorch

- Designed and implemented an **AutoEncoder model** using **PyTorch**, achieving a 30% reduction in noise for handwritten digits and improving recognition accuracy by 25%, enhancing image clarity for machine learning applications.

Pharmacy Database Management System

- Built a **full-stack Pharmacy Database Management System** with **CRUD functionality**, integrating a user-friendly frontend for seamless data management. Utilized **Python** and **PostgreSQL** to enhance database efficiency and streamline pharmaceutical record-keeping.

Fault Analysis of DC Renewable Microgrids using Machine Learning

- Engineered detection algorithms using **MATLAB** and **Simulink** comparing six machine learning models, achieving an average accuracy rate of 92% in diagnosing faults; directly contributed to improved operational efficiency by reducing downtime during testing phases.

EDUCATION

PES University, Bangalore | Bachelor of Technology

2019 - 2023

Major in Electrical and Electronics Engineering

Minor in Computer Science Engineering

PUBLICATIONS

[1] A. Singh, A. N. Manapure, M. Kaur, J. Badam and A. Thakre, "Centralized Power Control In Wireless Interference Channel with Imperfect CSI," 2023 IEEE 20th India Council International Conference (INDICON), Hyderabad, India, 2023, pp. 939-943, doi: 10.1109/INDICON59947.2023.10440855.